

Hamiltonian formalism of quantum field theories using Daubechies wavelet basis

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I. INTRODUCTION TO DAUBECHIES WAVELET BASIS

In this section, we briefly review the construction of the Daubechies wavelet basis and its properties, which will be used in the rest of the paper. We refer the reader to [?] for a comprehensive introduction to the subject of discrete wavelets.

The elements of basis are constructed from a single function $s(x)$, called the mother scaling function, which is defined by the refinement equation,

$$s(x) = \sum_{n=0}^{2K-1} h_n s(2x - n). \quad (1)$$

where h_n 's are called the filter coefficients. The refinement equation expresses the mother scaling function in terms of scaled and shifted versions of itself. This can be made explicit by defining the dyadic scaling operator, $\hat{D}$, and the translation operator, $\hat{T}$,

$$\hat{D}f(x) = \sqrt{2}f(2x), \quad \hat{T}g(x) = g(x - 1), \quad (2)$$

where, $f(x)$ and $g(x)$, are arbitrary real functions. The operators, $\hat{D}$ and $\hat{T}$, are unitary with respect to the norm,

$$(f, g) = \int f(x)g(x)dx \quad (3)$$

They do not commute with each other. Specifically,

$$\hat{T}\hat{D} = \hat{D}\hat{T}^2. \quad (4)$$

Fig. ?? shows the graphical view of the action of $\hat{D}$ and $\hat{T}$. In terms of operators, D and T , the form of the refinement equation,

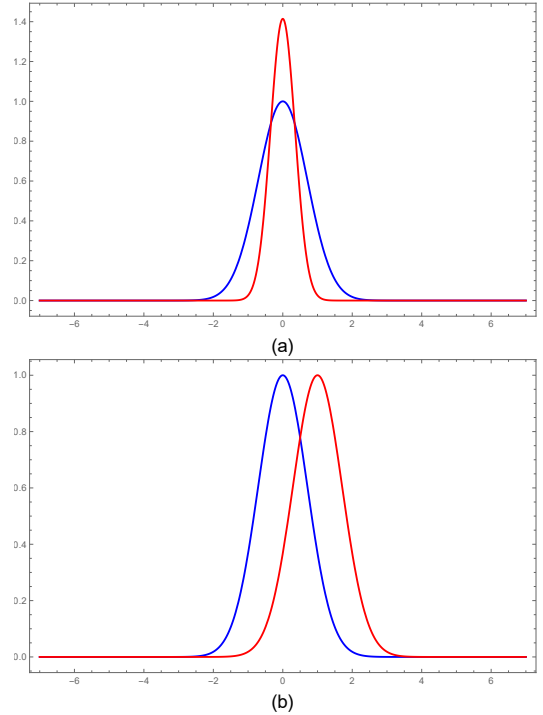
$$s(x) = \sum_{n=0}^{2K-1} h_n \hat{D}\hat{T}^n s(x), \quad (5)$$

shows that the mother scaling function of order K is generated through a specific linear combination of $2K$ number of translated and dyadically scaled replicas of itself. Eq. (??) only determines the relative values of the mother scaling at different points on the real line. The absolute values of the mother scaling function at different points are fixed by demanding that

$$\int s(x)dx = 1. \quad (6)$$

The procedure for determining the filter coefficients and the mother scaling function, $s(x)$ itself, starting from Eq. (??) and Eq. (??), is outlined in (cite references). In Fig. ??, we provide a visual representation of the mother scaling function and its associated refinement equation, Eq. (??) for $K = 3$.

FIG. 1. Any generic function $f(x)$ (blue color). The action of (a) $\hat{D}$ and (b) $\hat{T}$ operator on that function (red color)



The order K determines the extent of smoothness and support of the mother scaling function. For example, the mother scaling function is singly differentiable for $K = 2$ and doubly differentiable for $K = 3$. Its support is given by the interval $[0, 2K - 1]$.

The resolution k scaling basis functions are defined by,

$$s_n^k(x) := \hat{D}^k \hat{T}^n s(x), \quad (7)$$

with the location index n ranging over the set of integers. Fig. ?? shows order $K = 3$ Daubechies scaling functions for a few different resolutions and locations. We demand that they constitute an orthonormal set,

$$\int s_n^k(x) s_m^k(x) dx = \delta_{nm}. \quad (8)$$

The arbitrary linear combination of these functions

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FIG. 2. The red dotted line is the mother scaling function $s(x)$ for $K = 3$, formed with the linear combination of the weighted sum of 6 translated and scaled copies of itself.

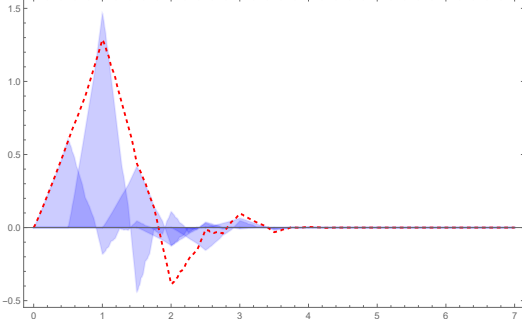
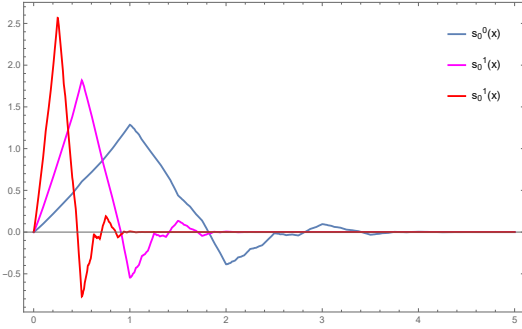


FIG. 3. Order, $K = 3$ Daubechies scaling functions for three different resolution $k = 0, 1$ and 2 respectively.



generates the approximation space $\mathcal{H}^k$ of resolution k ,

$$\mathcal{H}^k = \left\{ f(x) \mid f(x) = \sum_{n=-\infty}^{\infty} f_n^{s,k} s_n^k(x), \sum_n |f_n^{s,k}|^2 < \infty \right\}. \quad (9)$$

Using Eq. (??), Eq. (??) and Eq. (??), we can show that resolution $k-1$ scaling basis functions can be expressed as linear combinations of resolution k scaling basis functions,

$$s_n^{k-1}(x) = \sum_{m=2n}^{2n+2K-1} H_{nm} s_m^k(x), \quad (10)$$

where $H_{nm} = h_{m-2n}$. This implies that the resolution $k-1$ approximation space is contained within resolution k approximation space,

$$\mathcal{H}^{k-1} \subset \mathcal{H}^k, \quad (11)$$

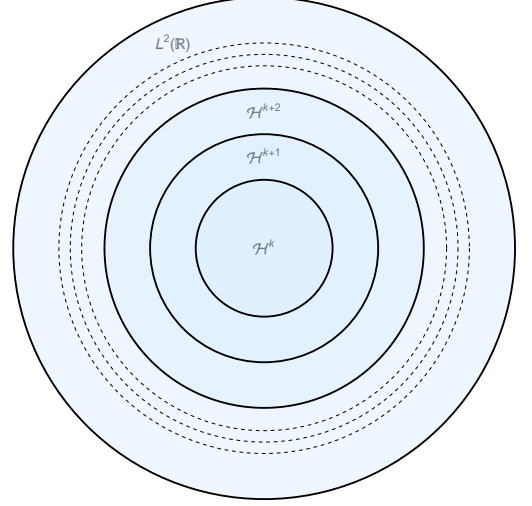
and by induction, for any $m > 0$

$$\mathcal{H}^k \subset \mathcal{H}^{k+m}. \quad (12)$$

Therefore, approximation spaces have a nested structure, as shown in Fig. ???. The infinite resolution limit of $\mathcal{H}^k$ generates the space of square-integrable functions $\mathcal{L}^2(\mathbb{R})$,

$$\mathcal{L}^2(\mathbb{R}) = \lim_{k \rightarrow \infty} \mathcal{H}^k. \quad (13)$$

FIG. 4. Construction of the space of the square integrable functions by taking infinite resolution limit of the scaling function space $\mathcal{H}^k$.



Since $\mathcal{H}^k$ is a proper subspace of $\mathcal{H}^{k+1}$, we can define $\mathcal{W}^k$ as the orthogonal complement of $\mathcal{H}^k$ in $\mathcal{H}^{k+1}$. $\mathcal{W}^k$ is referred to as the detail or wavelet space of resolution k . This implies that the resolution $k+1$ approximation space is a direct sum of resolution k , approximation and detail subspaces.

$$\mathcal{H}^{k+1} = \mathcal{H}^k \oplus \mathcal{W}^k. \quad (14)$$

To construct the basis functions that span $\mathcal{W}^k$, we define the mother wavelet function, $w(x)$,

$$w(x) := \sum_{n=0}^{2K-1} g_n \hat{D} \hat{T}^n s(x), \quad (15)$$

where the coefficients

$$g_n := (-1)^n h_{2K-1-n}, \quad (16)$$

have been chosen such that the mother wavelet function $w(x)$ is orthogonal to $s(x)$. The basis functions for space $\mathcal{W}^k$, referred to as the resolution k wavelet functions, are defined by,

$$w_n^k(x) := \hat{D}^k \hat{T}^n w(x). \quad (17)$$

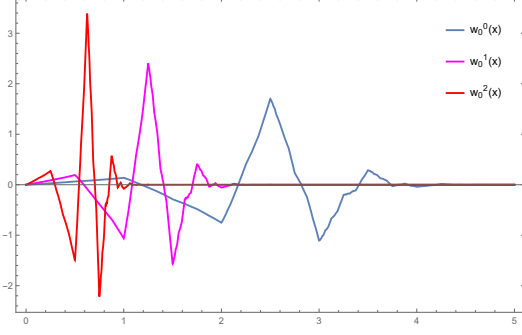
Fig. (??) shows order $K = 3$ Daubechies scaling functions for a few different resolutions and locations. By design, the wavelet functions constitute an orthonormal set,

$$\int w_n^k(x) w_m^l(x) dx = \delta_{nm} \delta_{kl}, \quad (18)$$

that are orthogonal to the scaling functions,

$$\int s_m^k(x) w_n^{l+k}(x) dx = 0, \quad l \geq 0, \quad (19)$$

FIG. 5. Order, $K = 3$ Daubechies wavelet functions for different resolutions $k = 0, 1$ and 2 respectively.



The resolution k wavelet functions span the resolution k wavelet space $\mathcal{W}^k$,

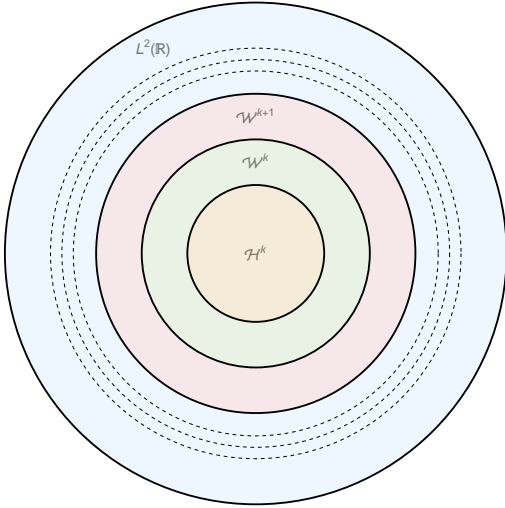
$$\mathcal{W}^k = \left\{ f(x) \mid f(x) = \sum_{n=-\infty}^{\infty} f_n^{w,k} w_n^k(x), \sum_n |f_n^{w,k}| < \infty \right\} \quad (20)$$

The space of square integrable functions, $\mathcal{L}^2(\mathbb{R})$, can be constructed by recursive use of Eq. (??).

$$\mathcal{L}^2(\mathbb{R}) = \mathcal{H}^k \oplus \mathcal{W}^k \oplus \mathcal{W}^{k+1} \oplus \mathcal{W}^{k+2} \dots, \quad (21)$$

as depicted in Fig. ??.

FIG. 6. Euler diagram of forming the space of square integrable functions ($\mathcal{L}^2(\mathbb{R})$) with scaling and wavelet basis.



There are two possible basis choices for $\mathcal{H}^k$: one can either use the scaling functions $\{s_n^k(x)\}_{n=-\infty}^{\infty}$ of resolution k or a combination of the scaling and wavelet functions of resolution $k-1$, $(\{s_n^{k-1}(x)\}_{n=-\infty}^{\infty} \cup \{w_n^{k-1}(x)\}_{n=-\infty}^{\infty})$. These two bases are related to each other through an orthogonal

transformation:

$$s_n^{k-1}(x) = \sum_{m=2n}^{2n+2K-1} H_{nm} s_m^k(x), \quad (22)$$

$$w_n^{k-1}(x) = \sum_{m=2n}^{2n+2K-1} G_{nm} s_m^k(x), \quad (23)$$

$$s_n^k(x) = \sum_{m=2n}^{2n+2K-1} H_{nm}^t s_m^{k-1}(x) + \sum_{m=2n}^{2n+2K-1} G_{nm}^t w_m^{k-1}(x), \quad (24)$$

where,

$$H_{nm} = h_{m-2n}, \quad G_{nm} = g_{m-2n} \quad (25)$$

$$H_{nm}^t = h_{n-2m}, \quad G_{nm}^t = g_{n-2m} \quad (26)$$

The h_n coefficients that are appearing in Eq. (??), can be determined for any integral value of K by solving the following system of equations:

$$\sum_{n=0}^{2K-1} h_n = \sqrt{2}, \quad (27)$$

$$\sum_{n=0}^{2K-1} h_n h_{n-2m} = \delta_{m0}, \quad (28)$$

$$\sum_{n=0}^{2K-1} n^m g_n = \sum_{n=0}^{2K-1} n^m (-1)^n h_{2K-1-n}, \quad (29)$$

$$= 0, \quad m < K.$$

Equation (??), can be found substituting the normalization condition of the scaling function, Eq (??), into the scaling equation, Eq. (??). Equation (??), can be found using the scaling equation, Eq. (??), and the orthonormality condition of the scaling functions, Eq. (??). Equation (??), ensures that the wavelet functions are orthogonal to degree $K-1$ polynomials.

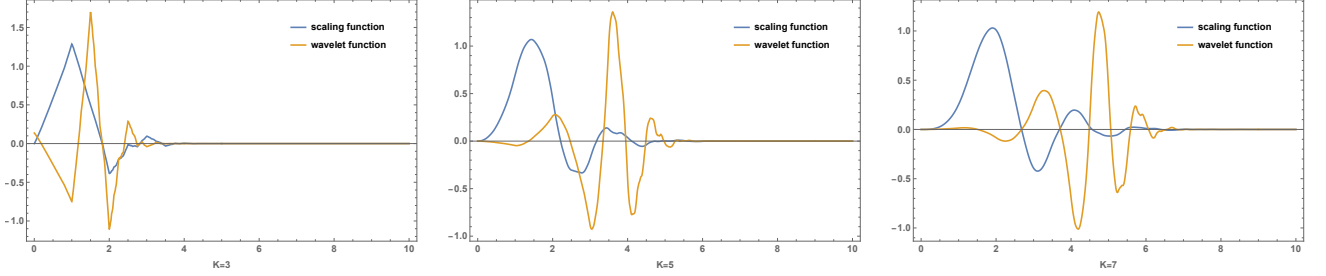
The h_n coefficients for $K = 1, 2, 3, 4$ and 5 are given in Table. ?. Utilizing the values of h_n , we can find out the values of scaling function, $s(x)$ and the wavelet function, $w(x)$ for each points of x recursively from Eq. (??) and Eq. (??). For any arbitrary value of K , the mother scaling function, $s(x)$ and the mother wavelet function, $w(x)$ will have the support within the interval $[0, 2K-1]$. The function $s_n^k(x)$ and $w_n^k(x)$ will have the compact support smaller by 2^k in comparison with the $s(x)$ and $w(x)$. The graphical view of the $s(x)$ and $w(x)$ for different values of $K = 2, 4$ and 6 is given in Fig. ??.

$$s_n^k(x), w_n^k(x) \neq 0 \quad \forall x \in \left(\frac{n-0}{2^k}, \frac{n+2K-1}{2^k} \right) \quad (30)$$

$$\implies \text{support size} = \frac{2K-1}{2^k}. \quad (31)$$

TABLE I. h coefficients of Daubechies wavelet for different values of K .

h_n	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 5$
h_0	0.707107	0.482963	0.332671	0.230378	0.160102
h_1	0.707107	0.836516	0.806892	0.714847	0.603829
h_2	0	0.224144	0.459878	0.630881	0.724309
h_3	0	-0.12941	-0.135011	-0.0279838	0.138428
h_4	0	0	-0.0854413	-0.187035	-0.242295
h_5	0	0	0.0352263	0.0308414	-0.0322449
h_6	0	0	0	0.032883	0.0775715
h_7	0	0	0	-0.0105974	-0.00624149
h_8	0	0	0	0	-0.0125808
h_9	0	0	0	0	3.33573×10^{-3}

FIG. 7. The plot of scaling and wavelet function for different values of K .

II. SCALAR FIELD THEORY IN WAVELET BASIS

In this section, to introduce the formulation of the wavelet-based field theory, we used the example of the free scalar field theory. We first review the conventions of the free scalar field theory, and then present the formulation of the theory within the wavelet-based framework. This introductory recipe can be applied to interacting field theories as well.

A. The scalar field theory: Conventions

The Lagrangian density of the free scalar field theory of mass m is given by,

$$\mathcal{L}_0 = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{m^2}{2} \phi^2. \quad (32)$$

The corresponding Hamiltonian can be written as,

$$H_0 = \int dx \frac{1}{2} \left(\pi^2 + (\nabla \phi)^2 + m^2 \phi^2 \right), \quad (33)$$

where, π is the conjugate momentum of the the field ϕ ,

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}. \quad (34)$$

The canonical conjugate variables, $(\phi(x, t), \pi(x, t))$, satisfy the following equal-time commutation relation,

$$[\phi(x, t), \pi(y, t)] = i\delta(x - y). \quad (35)$$

The field-operators can be expressed in terms of the Fourier modes as $(-+++ \text{ metric})$,

$$\phi(x) = \int \frac{dp}{\sqrt{2(2\pi)E_m(p)}} (e^{-ip \cdot x} a^\dagger(p) + e^{ip \cdot x} a(p)) \quad (36)$$

$$\pi(x) = \int dp \sqrt{\frac{E_m(p)}{2(2\pi)}} (e^{-ip \cdot x} a^\dagger(p) - e^{ip \cdot x} a(p)) \quad (37)$$

where, we use the shorthand $x \equiv (x, t)$ for all the space-time points, and $E_m(p)$ is the single particle energy,

$$E_m(p) = \sqrt{m^2 + p^2}. \quad (38)$$

The equal-time commutation relation between the field variables, Eq. (??), implies the following commutation relation between the operator-valued Fourier coefficients, $a(p)$ and $a^\dagger(p)$,

$$[a(p), a^\dagger(p')] = \delta(p - p'). \quad (39)$$

The momentum-space representation of the free field theory Hamiltonian is given by,

$$H_0 = \int dp E_m(p) a^\dagger(p) a(p). \quad (40)$$

This relation implies that the Fock-space basis states built from definite momentum modes diagonalize the free Hamiltonian. Next, we will see that the Fock-space constructed from the wavelet-modes does not consists of the eigenstates of the free scalar field theory Hamiltonian.

B. The scalar field theory: wavelet-based framework

In this subsection, we present the wavelet-based formalism of quantum field theory that will be used throughout the paper. We begin with the decomposition of the field operators as a linear combinations of the product of the basis functions and the time dependent operator coefficients as,

$$\begin{aligned}\phi(x) &= \lim_{k \rightarrow \infty} \sum_{n=-\infty}^{\infty} \phi_n^{s,k}(t) s_n^k(x), \\ \pi(x) &= \lim_{k \rightarrow \infty} \sum_{n=-\infty}^{\infty} \pi_n^{s,k}(t) s_n^k(x),\end{aligned}\quad (41)$$

where,

$$\phi_n^{s,k}(t) = \int dx \phi(x, t) s_n^k(x), \quad \pi_n^{s,k}(t) = \int dx \pi(x, t) s_n^k(x), \quad (42)$$

here, the scaling field coefficients, $\phi_n^{s,k}$ and $\pi_n^{s,k}$, describes the physics down to the length scale $\frac{2K-1}{2^k}$.

To represent the free scalar field Hamiltonian in terms of the operator-valued scaling coefficients, we substitute the field variables $\phi(x)$ and $\pi(x)$ in Eq. (??) with their corresponding expansions given in Eqs. (??) and obtain the following form of the Hamiltonian:

$$\begin{aligned}H_0 := \frac{1}{2} & \left(\sum_n \pi_n^{s,k}(t) \pi_n^{s,k}(t) + \sum_{m,n} \phi_m^{s,k}(t) \phi_n^{s,k}(t) D_{ss,mn}^k \right. \\ & \left. + \sum_n \mu^2 \phi_n^{s,k}(t) \phi_n^{s,k}(t) \right),\end{aligned}\quad (43)$$

is the projection of the Hamiltonian to the truncated subspace $\mathcal{H}^k$, describes the physics of the length scales down to $\frac{2K-1}{2^k}$.

The interaction between different field modes appearing in Eq. (??) denoted with $D_{ss,mn}^k$, is the overlap integral between the first-order derivative of the scaling functions, and is given by,

$$D_{ss,mn}^k = \int dx \frac{\partial s_m^k(x)}{\partial x} \frac{\partial s_n^k(x)}{\partial x}, \quad (44)$$

These overlap integrals can be analytically evaluated using the properties of the wavelet functions. The detail procedure is described in [1]. Due to the compact support of these functions the value of these integrals decreases with the increasing spatial separation (i.e $|m-n|$) between them. This signifies that the interaction strength between two different scaling modes decreases with the increasing distance between them.

The canonical commutation relation between the operator-valued scaling coefficients, defined in Eqs. (??) are given by,

$$\begin{aligned}[\phi_m^{s,k}(t), \pi_n^{s,k}(t)] &= i\delta_{mn}, \quad [\phi_m^{s,k}(t), \phi_n^{s,k}(t)] = 0, \\ [\pi_m^{s,k}(t), \pi_n^{s,k}(t)] &= 0,\end{aligned}\quad (45)$$

which are obtained from the usual commutation relation given in Eq. (??).

The full Hamiltonian leads to a compact matrix form in terms of these operator-valued mode variables,

$$H = \frac{1}{2} [\Pi^T \Pi + \Phi^T \Omega^2 \Phi], \quad (46)$$

where, Φ and Π are the column matrices containing the field and the conjugate momentum variables respectively,

$$\begin{aligned}\Phi^T &= (\phi_{-\infty}^{s,k} \quad \dots \quad \phi_m^{s,k} \quad \dots \quad \phi_{\infty}^{s,k}), \\ \Pi^T &= (\pi_{-\infty}^{s,k} \quad \dots \quad \pi_m^{s,k} \quad \dots \quad \pi_{\infty}^{s,k}),\end{aligned}\quad (47)$$

and the couplings between different scaling field variables are contained within the positive definite $\Omega^2 = \mu^2 \delta_{mn} + D_{ss,mn}^k$ matrix.

In this formulation, the free scalar field theory is reformulated as a theory of infinitely many linearly coupled harmonic oscillators, labeled by the resolution(k) and translation(m) indices. The coupling strength between these oscillators (with the the same/different resolution index), decreases with the increasing value of the spatial separation, measured by the difference between the translation indices associated with those oscillators, and exactly vanishes after a certain value of the separation. For example, in the case of order 3($K=3$) Daubechies wavelet functions the coupling between two oscillators is non-zero only for $|(m-n)| \leq 4$.

Form Eq. (??), we can define an annihilation and a creation operator column matrices as follows,

$$a = \frac{1}{\sqrt{2}} (A\Phi + iB\Pi), \quad a^\dagger = \frac{1}{\sqrt{2}} (A\Phi - iB\Pi), \quad (48)$$

where, $A = \sqrt{\Omega}$, $B = \frac{1}{\sqrt{\Omega}} = \frac{1}{A}$. Φ , and Π are the field operator column matrices given in the Eq. (??). Due to the positive definite nature of the Ω^2 -matrix, the two matrices A and B are well-defined.

The operator column matrices a and $a^\dagger$, given in Eq. (??), contains a set of annihilation and creation operators corresponding to the scaling mode given by,

$$\begin{aligned}a_m^{s,k} &:= \frac{1}{\sqrt{2}} \sum_n [A_{ss,mn} \phi_n^{s,k} + iB_{ss,mn} \phi_n^{s,k}], \\ a_m^{s,k\dagger} &:= \frac{1}{\sqrt{2}} \sum_n [A_{ss,mn} \phi_n^{s,k} - iB_{ss,mn} \phi_n^{s,k}],\end{aligned}\quad (49)$$

The inverse of these relations can be written down in the compact column matrix form as,

$$\begin{aligned}\Phi &= \frac{B}{\sqrt{2}} (a^\dagger + a) = \frac{1}{\sqrt{2\Omega}} (a^\dagger + a), \\ \Pi &= i \frac{A}{\sqrt{2}} (a^\dagger - a) = i \sqrt{\frac{\Omega}{2}} (a^\dagger - a).\end{aligned}\quad (50)$$

Substituting Φ and Π of Eq. (??) with the expression of these operators given in Eq. (??), we obtain the following

Hamiltonian,

$$H = (a^\dagger)^T \Omega a + \frac{1}{2} \text{Tr}[\Omega]. \quad (51)$$

As Ω is infinite in dimensions, the value of the second term of the right hand side of the above expression is infinity, which represents the zero-point energy of the system. The normal-ordered Hamiltonian can be

obtained by rescaling the zero point energy to zero by omitting the second term of the Hamiltonian. The normal-ordered Hamiltonian in terms of scaling and wavelet creation and annihilation operator can be written as,

$$H = \sum_{m,n,(l \geq k)} \Omega_{ss,mn}^k a_m^{s,k\dagger} a_n^{s,k}. \quad (52)$$

Now, we proceed construct the Fock space basis elements using the scaling modes.

$$S = \left| m1_{-\infty}^{s,k} \dots m1_p^{s,k} \dots m1_{\infty}^{s,k} \right\rangle. \quad (53)$$

Here, $m1_p^{s,k}$ denotes the the occupation numbers of the scaling mode with the translation and resolution indices p and k respectively.

The action of the creation and annihilation operators, given in Eq. (??), on these states are given by,

$$a_p^{s,k\dagger} \left| m1_{-\infty}^{s,k} \dots m1_{\infty}^{s,k} \right\rangle = \sqrt{m1_p^{s,k} + 1} \left| m1_{-\infty}^{s,k} \dots (m1_p^{s,k} + 1) \dots m1_{\infty}^{s,k} \right\rangle \quad (54)$$

$$a_p^{s,k} \left| m1_{-\infty}^{s,k} \dots m1_{\infty}^{s,k} \right\rangle = \sqrt{m1_p^{s,k}} \left| m1_{-\infty}^{s,k} \dots (m1_p^{s,k} - 1) \dots m1_{\infty}^{s,k} \right\rangle \quad (55)$$

After defining the action of the creation and the annihilation operators on the basis states, we construct the Hamiltonian matrix,

$$H_{0ss,m1m2} = \left\langle m1_{-\infty}^{s,k} \dots m1_{\infty}^{s,k} \right| H_0 \left| m2_{-\infty}^{s,k} \dots m2_{\infty}^{s,k} \right\rangle. \quad (56)$$

C. Truncation

The resulting Hamiltonian matrix is infinite in dimensions. To obtain a finite Hamiltonian matrix, the Hilbert space is restricted by introducing ultraviolet and infrared cutoffs. The ultraviolet cutoff is implemented by limiting the resolution of the basis functions, while the restriction of the translation index amounts to the physical spatial volume, which in turn provide the ultraviolet cutoff to the theory. Furthermore, the occupation number of each Fock basis states is limited to 10. The periodic boundary condition is applied to construct the Hamiltonian and the simulation box size is chosen to be $|l| \leq 10$. Since the Hamiltonian operator commutes with the momentum operator and the parity operator, the Hamiltonian can be arranged in block-diagonal form, with each block representing a sector of definite momentum and parity. To obtain the low-lying eigenvalues, we projected out the zero momentum-

even parity sector of the Hamiltonian, which drastically reduces the dimension of the Hamiltonian matrix. The procedure of projecting out the zero momentum-even parity block of the Hamiltonian matrix is presented in Appendix ???. Table. ??, presents a few low-lying the eigenvalues of the free scalar field theory for increasing resolutions.

As presented in the Table. ??, the eigenvalues are gradually converging towards the exact value with the increasing resolution. This indicates that, with the systematic increment of the ultraviolet cutoff improves the precision of the numerical estimate within the wavelet-based framework. This validates the effectiveness of the truncated wavelet-based Hamiltonian framework in accurately capturing the low-lying energy spectrum of the free scalar field theory. Having established the reliability of this framework, in the next section, we will now apply this formalism to the $1 + 1$ dimensional ϕ^4 theory to investigate its nonperturbative dynamics.

TABLE II. Comparison of exact vs. computed eigenvalues of ground state, first excited state, second excited state with increasing resolutions (k).

State	Exact	$k = -1$	$k = 0$	$k = 1$
$1p$	1	1.04363	1.01152	1.00408
$2p$	2	2.08726	2.02304	2.00815
$3p$	3	3.13093	3.03457	3.01223
$4p$	4	4.17466	4.04613	4.01630